Appendix B: Projects

Listed in this section of the text are some potential projects for individual students or teams of students to develop. The level of development will be a function of the number of people participating, the length of time available (10 week trimester, 15 week semester, etc.). The topics were compiled from many years of conducting these projects at RPI, and were selected due to the ease of purchasing parts.

If time permits, it has been found to be very useful for the student(s) to build a mockup of one or more facets of the prototype automation. This mockup can be made from materials such as wood rather than metal, and might be human powered rather than actuator driven. The key to choosing what to mock up lies in the needs to investigate a potential automation solution that has a fairly high risk factor, or is just an unknown. Experience shows that all of the CAD modeled designs in the world may not address these questions. After completion of the mockup, the student(s) and faculty would have a better sense that the proposed method would work.

Students are also encouraged to find their own project to develop. Faculty may be of great assistance also. All one has to do is to talk to local companies and offer a "what if" look at some automation, and the possible projects will seemingly fall out of the sky.

Whatever project is selected, it is highly recommended that there be sample products with which to work. Professional automation designers always insist on a few samples, both as loose parts and as a final product, before they will

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brainstorm an answer for developing a proposal. So student teams should be encouraged to purchase a few parts where it is logical, if they are not provided free by local companies looking for assistance.

PROJECT NUMBER AND NAME: 1, ASSEMBLE DOOR HINGES AND PACKAGE

- *Task* — Assemble standard door hinges from the two halves, and insert the pin. Care needs to be taken to keep pin in place while holding in place under vacuum head that seals warm plastic to printed cardboard backing material.
- *Recommended Production Rate* — 12 hinges per minute.
- *Recommended Items to Purchase* — Several door hinges on vacuum-sealed package. Pin must be able to be removed.
- *Possible Mockup* — Model hinge alignment and pin insertion.

PROJECT NUMBER AND NAME: 2, HANDLE, COUNT, AND PACKAGE FLEXIBLE DRINKING STRAWS

- *Task* — To accumulate, count, and place into a box 100 plastic drinking straws. Straws need to be handled in a relatively clean environment since they are used directly out of the box by consumers. Straws may be straight, or they may have a section capable of bending. System should work for both.
- *Recommended Production Rate* — Box or 100 straws should be filled every two seconds.
- *Recommended Items to Purchase* — Several boxes of straws. May need boxes from other products.
- *Possible Mockup* — Hand-powered singulation and counting station.

PROJECT NUMBER AND NAME: 3, DISPENSE PALLETIZING SLIP SHEET

- *Task* — A slip sheet is a piece of standard paper placed between layers of boxes on a pallet to create better structural rigidity. Paper can be as simple as newsprint. Paper can either come from a continuous roll or a stack of sheets. Student team is to select which. Automation needs to be very quick since this is a time where no boxes get placed on pallet. Rest of motion can be slower while a new layer is placed.
- *Recommended Production Rate* — One sheet placed in one second.
- *Recommended Items to Purchase* — Large drawing pad of artists paper.

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- *Possible Mockup* — Hand-powered gripper to grab paper from roll or stack.

PROJECT NUMBER AND NAME: 4, PALLET DISPENSER

- *Task* — Pallets need to be placed at base of palletizer. Pallets can be wood (rough surface) or plastic (relatively smooth). Empty pallets will be stacked up to 4 ft high. Pallet to be loaded can be grabbed from top of the stack or the bottom.
- *Recommended Production Rate* — Pallet needs to be dispensed into working area in two seconds.
- *Recommended Items to Purchase* — Several wooden pallets. Can usually be obtained for free at larger supermarkets and department stores. Wooden pallets often burned rather than recycled.
- *Possible Mockup* — Hand-powered station to move one pallet from stack. Can create a scaled version.

PROJECT NUMBER AND NAME: 5, HANDLE BOTTLES OF SPICES AND RELABEL

- *Task* — Custom label needs to be placed on generically labeled bottle of spice. New label must cover the original smaller label. New label is self-adhesive type. Bottles of spices are packed 24 to a cardboard case. Bottles need to be removed from case, new labels attached, and repackaged into a similar cardboard case.
- *Recommended Production Rate* — One case per two minutes.
- *Recommended Items to Purchase* — Several bottles of spice. The type found in dollar stores is best. Find preprinted labels or stickers.
- *Possible Mockup* — Create a station to apply labels and attempt to achieve a smooth and aligned application.

PROJECT NUMBER AND NAME: 6, PIZZA STACKING SYSTEM

- *Task* — Stack six frozen pizzas, each in vacuum-type plastic wrap, into a box. Stacking process can be done pizza right side up or upside down, but if upside down then the box needs to be filled from the bottom. Pizzas need to be transported right side up. Box flaps need to be closed and taped.
- *Recommended Production Rate* — One box of six pizzas per ten seconds.

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- *Recommended Items to Purchase* — Several frozen pizzas that are vacuum plastic wrapped.
- *Possible Mockup* — Material handing method to move pizzas with their irregular surface shape.

PROJECT NUMBER AND NAME: 7, ELECTRICAL WIRE STRIPPER AND BENDER MACHINE

- *Task* — 110 V and 220 V circuits are pre-wired into a circuit breaker panel. The wire is fairly stiff and the lengths are calculated by electricians. Wire from a spool needs to straightened, bent at 90°, and stripped at both ends.
- *Recommended Production Rate* — One stripped wire with two 90° bends every 5 seconds.
- *Recommended Items to Purchase* — Roll of #12 single conductor wire and several terminal strips to emulate circuit breakers.
- *Possible Mockup* — Hand-operated station to bend wire and strip one end.

PROJECT NUMBER AND NAME: 8, BOOK PICKING

- *Task* — E commerce companies like Amazon.com have thousands of pallets filled with books. Automation needs to be able to find the top book on a pallet and grab it without damaging the loose book jacket. The book needs to keep some identity while it gets transported to order processing location. Book location on the pallet is in interlocking rows.
- *Recommended Production Rate* — Grab one book from pallet and place onto transport system in five seconds.
- *Recommended Items to Purchase* — Several cheap hardcover books with jacket.
- *Possible Mockup* — Book grabbing device.

PROJECT NUMBER AND NAME: 9, BOOK PACKAGING

- *Task* — This Project takes the output of Project 8 and by using a database, places the order in the smallest box possible. Project should consider three different sizes of boxes, and a limit of six books. Boxes would need to be filled with packing materials (peanuts, etc.) to limit damage, and then closed and sealed.

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- *Recommended Production Rate* — Fill a box of six books and close in 30 seconds.
- *Recommended Items to Purchase* — Several cheap hardcover books with jacket.
- *Possible Mockup* — Book transfer station so as not to drop in books and damage them.

PROJECT NUMBER AND NAME: 10, UNLOAD AND FLIP COFFEE CAKES AND STACK PANS

- *Task* — Coffee cakes are baked in a ring or Bundt pan, and cooked in a continuous oven. Cakes then go through a cooling station and must be unloaded by flipping them upside down. They must be placed on a stiff cardboard sheet within 1/2 in. of center. The empty pans are then to be stacked, with a limit of stack height of 3 ft.
- *Recommended Production Rate* — 16 per minute.
- *Recommended Items to Purchase* — Several ring or Bundt pans and several cake mixes.
- *Possible Mockup* — Hand-powered cake-flipping station.

PROJECT NUMBER AND NAME: 11, LOW-COST REMOTE PALLET STRETCH WRAPPER

- *Task* — To wrap full pallets of mixed goods with stretch wrap while the pallet is on a human-steered forklift truck. Current stretch wrappers require full pallet to return to home base, but mixed goods pallet might come loose getting there. So pallet needs to be wrapped as soon as it is filled by last goods on top. Starting the wrapping is a major concern.
- *Recommended Production Rate* — Wrap 5 ft high pallet with two or more wrap layers in 45 seconds.
- *Recommended Items to Purchase* — Several boxes of Saran wrap or generic equivalent.
- *Possible Mockup* — Scale model using Saran wrap and small pallet of mixed goods. Hand powered possibly.

PROJECT NUMBER AND NAME: 12, SCOOP ICE CREAM CONE FROM FREEZER

- *Task* — To make ice cream cones at dairy store without human intervention. Should work with larger freezer with sliding doors to open

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the top. Ice cream is in square three gallon cardboard containers, stored in three rows of eight flavors each. Cones are square bottom type. Single scoop sized cones only. Gripper needs to be rinsed after each made cone.

- *Recommended Production Rate* — One cone every 20 seconds.
- *Recommended Items to Purchase* — Several different types of ice cream scoops and a few half gallons of ice cream.
- *Possible Mockup* — Hand-powered but designed for automation scoop.

PROJECT NUMBER AND NAME: 13, PACKAGE HEARING AID BATTERIES INTO VARIOUS PATTERNS

- *Task* — Take a single stream of hearing aid (calculator type) batteries and place them into several different package layouts. Placement is to be into the clear molded plastic part of a package, and can either be in a row or a circular pattern. Batteries cannot be mishandled significantly or they may be damaged or shorted out.
- *Recommended Production Rate* — Average eight batteries per four second cycle.
- *Recommended Items to Purchase* — Several different styles of packaged batteries.
- *Possible Mockup* — Create gripper device to transfer batteries into different patterns.

PROJECT NUMBER AND NAME: 14, TRANSFER PETRI DISHES FROM STORAGE TO AN INSPECTION SYSTEM

- *Task* — A stack of 3 in. diameter Petri dishes with attached covers need to be unstacked and manipulated so as to pass under a vision camera for inspection. The stack can be 25 dishes high, and each Petri dish is 0.75 in. in height. The stack of inspected dishes needs to be in the same order as the starting stack.
- *Recommended Production Rate* — One dish per five seconds through inspection station.
- *Recommended Items to Purchase* — Several Petri dishes from websites.
- *Possible Mockup* — Method to remove dish from stack as it gets shorter, or method to remove bottom dish from stack without stack falling over.

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PROJECT NUMBER AND NAME: 15, REMOVE WRAP FROM BOTH SIDES OF PLASTIC SHEET

- *Task* — Polycarbonate sheet plastic is often used instead of glass in windows. Pieces over 1 in. thick are bullet proof. Sheets come with thin plastic film on each side to protect from scratches, and this film must be removed before use. The film is held on by what appears to be significant static attraction, although this is only a guess. Film cannot be harshly removed by screwdriver or scratches appear.
- *Recommended Production Rate* — Remove film from both sides of 1 ft $\times$ 1 ft sheet in five seconds. Finding film edge is most difficult part.
- *Recommended Items to Purchase* — Some samples of Polycarbonate (Lexan) from hardware store.
- *Possible Mockup* — Film starting device for one side.

PROJECT NUMBER AND NAME: 16, LOAD SIX DONUTS FLAT INTO BOX

- *Task* — Six donuts are placed into a box, with a second row of three placed on top of the first row a of three. Donuts are round in shape, and are not wrapped individually in plastic. Boxes are pre-formed and the lid must be closed after filling.
- *Recommended Production Rate* — One box of six donuts every five seconds.
- *Recommended Items to Purchase* — Several boxes of donuts.
- *Possible Mockup* — Hand-powered donut placement station.

PROJECT NUMBER AND NAME: 17, STACK AND BOX FROZEN COOKIE DOUGH

- *Task* — Chocolate chip cookies are frozen in ready-to-bake form. The cookies are fairly large (2 in. diameter) and flat. Baking does not make them significantly larger in diameter as opposed to home baking results. Cookies are stacked in three rows by four columns, six cookies high in each box. Stacks of cookies can tilt significantly due to chip location variation.
- *Recommended Production Rate* — One box full of cookies every five seconds. Assume cookies are produced in rows of four.
- *Recommended Items to Purchase* — Several dozen already baked cookies and several boxes.
- *Possible Mockup* — Cookie stacking station and box loading station.

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PROJECT NUMBER AND NAME: 18, BREAD BAGGING

- *Task* — Standard 1 lb loaves of Italian bread have significant shape variation. Loaves are to be bagged in individual paper open end bags, and the ends twisted closed. Loaves are coming from a cooling conveyor in random position and orientation.
- *Recommended Production Rate* — One loaf bagged every two seconds.
- *Recommended Items to Purchase* — Eight to ten loaves of Italian bread in their paper bags.
- *Possible Mockup* — Model bread loaves and determining their position and orientation.

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